If we view the map $\mathscr{F}_t$ as a change of variables, we have

$$\mathrm{Vol}(\mathcal{S}(t)) = \int_{\mathcal{S}} |D| \mathrm{d}\omega,$$

where $D = \det\left(\frac{\partial \mathscr{F}_t}{\partial z}\right)$.

To understand where Liouville's theorem comes from, recall that the variational equations of the last chapter are a system of ordinary differential equations for $\boldsymbol{W}(t) = \mathscr{F}_t'(z(t))$:

$$\frac{\mathrm{d}\boldsymbol{W}}{\mathrm{d}t} = \boldsymbol{f}'(z(t))\boldsymbol{W}.$$

Thus

$$\dot{\boldsymbol{W}}\boldsymbol{W}^{-1} = \boldsymbol{f}'(z(t)).$$

Now let $D = \det(\boldsymbol{W})$. One can show (see Exercise 5) that

$$\frac{\dot{D}}{D} = \mathrm{tr}\left(\dot{\boldsymbol{W}}\boldsymbol{W}^{-1}\right).$$

This implies that

$$\dot{D} = \mathrm{div}(\boldsymbol{f}(z(t)))D,$$

and thus

$$D(t) = D(0)e^{\int_0^t \mathrm{div}(\boldsymbol{f}(z(s)))\mathrm{d}s}.$$

In particular, if $\mathrm{div}\boldsymbol{f} \equiv 0$, we see that $D \equiv D(0) = 1$ and it follows that the volume is constant.

Liouville's theorem may be summarized compactly as:

$$\nabla \cdot \boldsymbol{f} = 0 \Rightarrow \det \mathscr{F}_t' = 1.$$

Example 2.3 A 1-d oscillator with Lennard-Jones potential is described by the equations

$$\begin{aligned} \dot{q} &= p, \\ \dot{p} &= -\varphi_{\mathrm{LJ}}'(q). \end{aligned}$$